

Article

Early-Career Choices in the Era of Generative AI: Human Capital Depreciation, Career Anxiety, and Occupational Adaptation Among Fresh Graduates

Xinyu Cai ¹ and Tiantian Mo ^{1,*}

Academic Editor: Firstname Lastname

Received: date

Revised: date

Accepted: date

Published: date

Citation: To be added by editorial staff during production.

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¹ College of Business, Jiaying University, Jiaying 314001, China; caixinyu@zjxu.edu.cn* Correspondence: 00008227@zjxu.edu.cn

Highlights

What are the main findings?

- AI-related career anxiety converts into occupational adaptation along an inverted U-shaped path: adaptation peaks 0.37 standard deviations above mean anxiety, 38.5 percent of fresh graduates already stand beyond that turning point, and career avoidance rises monotonically with anxiety.
- Fuzzy-set qualitative comparative analysis identifies a single sufficient configuration for high adaptation — the conjunction of anxiety, generative-AI literacy, and employability support (consistency 0.892) — while no single condition is necessary and anxiety alone never suffices.

What are the implications of the main findings?

- Treating career anxiety as the error signal of an adaptive feedback loop, rather than as a hazard to be reduced, extends systems practice to individual-level workforce transformation: the loop's transmission, saturation, and conjunctural logic are each estimated rather than assumed.
- Universities, employers, and policymakers should pair any communication of AI-related threat with tool access, training time, and visible transition infrastructure, because capability and support — not reassurance — determine whether a record graduate cohort's anxiety becomes adaptation or avoidance.

Abstract

As generative artificial intelligence (GenAI) becomes most capable at precisely the entry-level tasks from which early careers have traditionally been built, fresh graduates must make their first occupational choices under conditions no previous cohort has faced. Drawing on socio-technical systems theory and a cybernetic control perspective, this study investigates how graduates' perceived human capital depreciation translates into AI-related career anxiety and, in turn, into occupational adaptation or career avoidance, and which combinations of conditions govern this process. Using a two-wave survey of 1,247 Chinese fresh graduates within three years of graduation (914 matched across waves, with behavioral outcomes measured six months after their antecedents), we employ confirmatory factor analysis, bootstrapped conditional-process models, and fuzzy-set qualitative comparative analysis (fsQCA). The study finds that (1) perceived human capital depreciation significantly raises AI-related career anxiety ($\beta = 0.487$), and perceived employability support buffers this first stage, reducing the pass-through from 0.64 to 0.33; (2) career anxiety exhibits an inverted U-shaped relationship with occupational adaptation, with a turning point 0.37 standard deviations above the mean anxiety level, beyond which 38.5 percent of graduates already fall, whereas career avoidance rises monotonically; (3) GenAI literacy displaces the turning point outward from 0.07 to 0.66 standard deviations, extending the range over which anxiety remains productive; and (4) the fsQCA identifies a single sufficient configuration for high adaptation, the conjunction of anxiety, literacy, and support (consistency 0.892), with no necessary condition and no consistent configuration for its negation, so anxiety alone never suffices. This study deepens the systems understanding of GenAI-era workforce development by modeling career anxiety as the error signal of an adaptive feedback loop, and offers practical guidance for universities, employers, and policymakers seeking to convert a record cohort's anxiety into adaptation rather than avoidance.

Keywords: generative artificial intelligence; career anxiety; human capital depreciation; occupational adaptation; socio-technical systems

1. Introduction

Generative artificial intelligence is reshaping the entry point of professional labor markets. The technology performs best on the codifiable, language-intensive, and well-specified tasks that have traditionally been assigned to the newest members of the professions [1–3]. Recent evidence shows entry positions contracting where the technology is adopted: junior-heavy hiring falls in exposed occupations [4,5], the productivity gains of the technology accrue disproportionately to novices while the demand for novices weakens [6,7], and employers increasingly describe entry roles in terms of supervising machine output rather than producing it [8,9]. For the roughly twelve million students who now graduate from Chinese higher education each year [10,11], these developments define the conditions under which the first occupation, the first investments in skill, and the first revisions of both must be chosen.

However, the existing literature has examined this collision almost exclusively from the demand side of the labor market [12–14]. Much less is known about the supply side: how the entrants themselves perceive the

depreciation of their just-acquired human capital, what that perception does to them psychologically, and when the resulting emotion converts into behavior that improves their position rather than behavior that worsens it. The question is of considerable practical importance, because early-career scarring is among the most persistent phenomena in labor economics: cohorts that stumble at entry carry lower earnings and worse matches for a decade or more [15–18]. If the first cohort of the GenAI era responds to depreciation signals with paralysis rather than adaptation, the technology's distributional effects will be amplified by the psychology of those it displaces.

The emerging research on artificial-intelligence anxiety has begun to document how workers feel about intelligent technologies [19–22]. Nevertheless, three gaps remain. Firstly, most studies treat anxiety as an outcome to be predicted or a hazard to be reduced, and therefore estimate only monotonic relationships between anxiety and behavior; whether anxiety can mobilize adaptive behavior within one range and suppress it within another has not been tested. Secondly, anxiety, capability, and institutional support have been studied separately, so it is unknown which combinations of these conditions are jointly sufficient for adaptive behavior. Thirdly, the population that matters most for workforce transformation — fresh graduates making their first occupational choices — has rarely been examined, even though students' perceptions of generative technologies are documented to be intense and ambivalent [23].

To address these gaps, this study adopts an explicitly systems-theoretic position. Drawing on socio-technical systems theory [24–26] and the cybernetic tradition [27–29], the fresh graduate is modeled as an adaptive agent inside a socio-technical labor system: the perception of human capital depreciation is the input signal, AI-related career anxiety is the error signal of a control loop that compares the career one expected with the career one is likely to obtain, and occupational adaptation or career avoidance is the behavioral output through which the loop closes. Three testable properties follow from this framing. A control signal should transmit the input (mediation); an error signal that saturates should stop producing corrective action at high amplitude (nonlinearity); and the corrective action should depend on the actuators available to the agent, namely capability and support (conjunction). The empirical design of this study maps directly onto these three properties.

This study employs a two-wave questionnaire survey of 1,247 Chinese fresh graduates within thirty-six months of graduation, with psychological antecedents measured at the first wave and behavioral outcomes measured six months later, and combines confirmatory factor analysis, bootstrapped conditional-process models with an explicitly curvilinear second stage [30–33], and fuzzy-set qualitative comparative analysis [34–36]. The research conclusions can help universities, employers, and policymakers recognize that the conversion of anxiety into adaptation depends on capability and support rather than on the level of anxiety alone, and provide both theoretical and practical guidance for workforce development in the GenAI era [37,38].

The remainder of this study is structured as follows: Section 2 elaborates on the theoretical background and the development of hypotheses; Section 3 outlines the research design; Section 4 presents the results; and Section 5 concludes.

2. Theoretical Background and Hypotheses Development

2.1. Perceived Human Capital Depreciation and AI-Related Career Anxiety

Human capital depreciates when the market value of a skill stock falls, whether through wear, through vintage effects, or through technological displacement of the tasks the skills perform [39–41]. The loss lands hardest on labor market entrants, because early careers are built by doing: skills are accumulated on the job [42], weighted and priced by the tasks a position offers [43], and advanced through internal ladders whose first rung is exactly the work that generative systems now absorb [44,45]. What matters for behavior, however, is not depreciation itself but its perception. A graduate observes the capability of generative systems in their own domain, compares it with what they were trained to do, and forms a judgment about how fast that training is losing value [21,23].

Appraisal theory holds that a perceived threat to a central resource elicits a corresponding negative activation state [46]. Conservation of resources theory makes the same prediction from a different direction: anticipated resource loss is the most potent known elicitor of stress, and the career is among the most central resources a labor market entrant holds [47,48]. From the cybernetic perspective, the depreciation percept is a widening discrepancy between the expected and the likely career trajectory, and the affective correlate of that discrepancy is anxiety [29]. Prior empirical work confirms that awareness of intelligent technologies is associated with career-related strain among incumbent workers [19–21] and that career anxiety is a distinct and measurable state among students and graduates [49]. Therefore, the following hypothesis is proposed:

H1. *Perceived human capital depreciation is positively associated with AI-related career anxiety.*

2.2. The Dual Behavioral Responses to Career Anxiety

Anxiety is an activation state, and activation is behaviorally ambidextrous. The coping literature distinguishes problem-focused responses, which attack the stressor, from avoidant responses, which attack the feeling [46]. In the career domain, the problem-focused response is occupational adaptation: enrolling in structured reskilling, pursuing credentials, deliberately practicing the new technology, and repositioning toward tasks the technology complements [50,51]. The avoidant response is career avoidance: postponing career decisions, avoiding information about the technology, and contemplating exit from the chosen field [20,21].

The relationship between anxiety and adaptation should not be monotonic. At low-to-moderate levels, anxiety supplies the activation that overcomes the inertia of costly investment; this mobilizing arm is among the oldest results in behavioral research [52] and is consistent with conservation of resources theory, in which moderate loss threat triggers resource investment [48]. At high levels, three mechanisms bend the curve downward. Cognitively, intense anxiety narrows attention and consumes the working memory that planning a retraining program requires. Motivationally, extreme threat triggers defensive avoidance rather than approach [46]. In resource terms, graduates deep in anticipated loss enter a defensive posture in which conserving what remains dominates investing in what could be [48]. The general form of this argument — a productive input becoming

counterproductive past an interior optimum — is the too-much-of-a-good-thing effect [53,54], and in control-theoretic language it is saturation: an error signal beyond the actuator's range ceases to improve regulation [28,29]. The avoidant response carries no interior optimum; whatever anxiety remains unconverted into action discharges as escape, monotonically. Therefore, the following hypotheses are proposed:

H2. *AI-related career anxiety has an inverted U-shaped relationship with occupational adaptation: adaptation rises with anxiety up to a turning point and declines beyond it.*

H3. *AI-related career anxiety is positively and monotonically associated with career avoidance.*

2.3. The Mediating Role of AI-Related Career Anxiety

If anxiety is the error signal through which the depreciation percept reaches behavior, then perceived depreciation should influence both behavioral responses through anxiety. Moreover, the transmitted effect should inherit the curvature of the second stage: because the local slope of the anxiety-adaptation curve changes sign at the turning point, the same depreciation signal that mobilizes a calm graduate should demobilize an already anxious one [30,31]. This conditional mediation is a stronger and more falsifiable claim than simple mediation, and it is the signature prediction of the control-loop framing. Therefore, the following hypotheses are proposed:

H4a. *AI-related career anxiety mediates the relationship between perceived human capital depreciation and occupational adaptation, and the indirect effect declines with the level of anxiety, turning negative beyond the turning point.*

H4b. *AI-related career anxiety mediates a positive relationship between perceived human capital depreciation and career avoidance.*

2.4. The Moderating Role of Generative-AI Literacy

Whether anxiety converts into adaptation or into escape depends on the actions available to the graduate. The first boundary condition is generative-AI literacy: the self-efficacious command of the technology itself, including prompting effectively, verifying machine output, and integrating it into one's own work [19,23,55]. Literacy raises the expected return on every unit of adaptive effort, because a graduate who can already use the tool converts a reskilling course into complementarity faster than one who cannot [56,57]. In the geometry of an inverted U, a higher payoff to effort holds the upward arm positive over a longer range and therefore displaces the turning point outward [33]. The same command of the tool should also drain the avoidance channel, because escape loses its appeal when engagement is feasible. Therefore, the following hypotheses are proposed:

H5a. *Generative-AI literacy moderates the inverted U-shaped relationship between anxiety and adaptation, displacing the turning point outward.*

H5b. *Generative-AI literacy weakens the positive relationship between anxiety and career avoidance.*

2.5. The Moderating Role of Perceived Employability Support

The second boundary condition operates one stage earlier in the loop. Perceived employability support is the graduate's assessment that university

career services, employer training provision, and their professional network stand behind their transition [58,59]. According to the buffering position of the stress literature, support reframes a threat as a solvable problem, so the same objective signal generates less strain in the first place [46,48]. In control-theoretic terms, support recalibrates the sensor rather than the actuator: it lowers the gain with which perceived depreciation is converted into anxiety. Therefore, the following hypothesis is proposed:

H6. *Perceived employability support weakens the positive relationship between perceived human capital depreciation and AI-related career anxiety.*

2.6. A Configurational Perspective on Adaptation

Interaction terms test whether conditions modify each other at the margin; they do not test the stronger systems claim that outcomes are produced by configurations — combinations of conditions that succeed or fail as wholes [35,60,61]. Complex adaptive systems typically exhibit equifinality, meaning multiple distinct paths lead to the same outcome, and configurational studies in the management and information-systems traditions routinely report several sufficient recipes [34,36,62,63]. The control-loop framing of this study, however, points the other way. If adaptation requires a signal (anxiety at a workable level), an actuator (literacy), and an enabling environment (support) simultaneously, then the sufficient recipes should collapse toward a single conjunction, and no individual ingredient should be sufficient or strictly necessary on its own. The configurational analysis therefore tests two claims against each other:

H7a. *No single psychological or contextual condition is necessary for high occupational adaptation.*

H7b. *Multiple distinct configurations of depreciation, anxiety, literacy, support, and exposure are sufficient for high occupational adaptation (equifinality).*

To summarize, the theoretical framework of this study is illustrated in Figure 1.

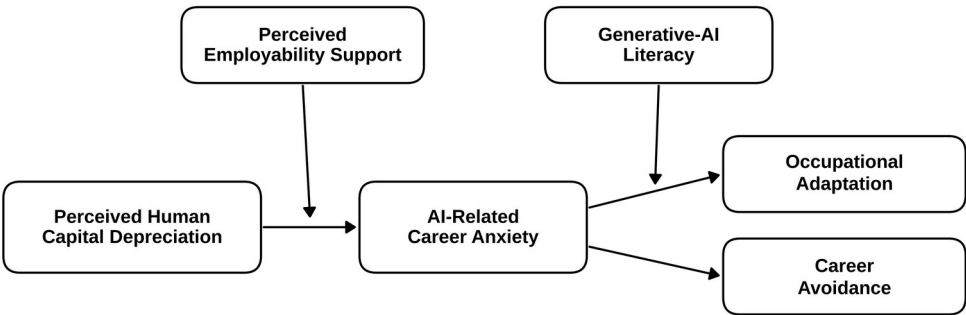


Figure 1. Theoretical research framework.

3. Data and Methods

3.1. Data Collection and Sample

This study takes mainland Chinese university graduates within thirty-six months of graduation as the targeted research sample, the population whose first occupational choices coincide with the diffusion of generative artificial

intelligence. To ensure the accuracy of the survey data, we developed the questionnaire by adapting established scales from the literature and incorporating expert suggestions, translated each item into Chinese following the standard back-translation procedure, and optimized the wording so that respondents could easily understand it. Respondents were recruited through a professional online panel supplemented by university alumni associations across provinces, with quotas on gender, degree level, and city tier to approximate the national graduate profile [10,64]. The first wave, fielded in September 2025, measured perceptions, psychological states, moderators, and controls; the second wave, fielded six months later in March 2026, measured the two behavioral outcomes. Attention checks, minimum-duration screens, and straight-lining filters were applied. A total of 1,247 valid first-wave responses were obtained, of which 914 were matched at the second wave, a retention rate of 73.3 percent. All statistical analyses were performed with documented code whose random seeds are fixed in the released scripts.

To ensure the sample is sufficiently representative and to rule out attrition bias, a sample structure and attrition analysis was conducted before the main analysis, as shown in Table 1. The retained and lost respondents do not differ significantly on any observable characteristic (the smallest p-value is 0.095), and anxiety itself does not predict attrition ($p = 0.479$); inverse-probability weighting is nevertheless applied in the robustness tests. The median monthly income of employed respondents is 6,300 yuan, in line with published graduate salary distributions [10]. Respondents span 20 occupation groups.

Table 1. Sample profile and attrition analysis.

Characteristic	Time 1 (N = 1247)	Retained (N = 914)	Lost (N = 333)	p
Female (%)	52.6	52.8	52.0	0.780
Age (years)	23.6	23.6	23.6	0.461
Master's degree (%)	18.2	18.4	17.7	0.787
STEM major (%)	40.4	40.3	40.8	0.854
Tier-1 city (%)	31.9	32.6	30.0	0.384
Months since graduation	18.7	18.7	18.6	0.850
Employed (%)	72.7	74.0	69.1	0.095
Parent with tertiary education (%)	28.4	28.0	29.4	0.626
Internship experience (%)	65.0	65.3	64.3	0.731
Depreciation items, mean	4.0	4.0	4.0	0.534
Anxiety items, mean	4.0	3.9	4.0	0.479

Note: Continuous characteristics are means; binary characteristics are percentages. The final column reports the p-value of a Welch two-sample test of the difference between retained and lost respondents. Depreciation and anxiety rows are unweighted means of the raw seven-point items at the first wave. Source: authors' own calculations.

3.2. Measurement

This study uses established questionnaire scales from the literature with appropriate modifications, using a seven-point Likert scale (1 = strongly disagree, 7 = strongly agree). The specific variables were measured as follows:

Perceived human capital depreciation (PHCD). Adapting skill-obsolescence measures [40,41] to the generative-AI context, four items capture the judgment that one's degree skills are losing labor market value as

generative systems improve (sample item: “The skills I acquired in my degree are losing value as generative AI systems improve”).

AI-related career anxiety (ANX). Drawing mainly on the artificial-intelligence anxiety scale of Wang and Wang (2022) [19], four items adapt the job-replacement dimension to the career frame of reference (“Thinking about what AI means for my career path makes me tense”).

Generative-AI literacy (LIT). Following the self-efficacy tradition [55] and studies of student GenAI competence [23], four items measure self-efficacious command of the technology: prompting effectively, verifying output, integrating it into one’s own work, and keeping pace with new releases.

Perceived employability support (SUP). Drawing on the employability literature [58], four items measure the graduate’s assessment of university career services, employer training provision, and network support.

Occupational adaptation (ADP). Measured at the second wave, four items record behavior over the intervening six months: enrollment in structured reskilling, credential pursuit, deliberate practice with generative tools, and movement toward tasks that complement the technology [50,51]. As a behavioral cross-check, the second wave also records hours of structured reskilling.

Career avoidance (AVD). Measured at the second wave, three items record postponed career decisions, avoidance of AI-related information, and contemplated exit from the field [20].

Occupational exposure (EXPO). The one non-perceptual condition: each respondent’s current or target occupation group is scored for generative-AI exposure by aggregating task-level language-model applicability to the occupation level, following the construction logic of the exposure literature [1,2], and standardized over the sample (raw-score mean 0.58, standard deviation 0.18).

Control variables. Gender, age, degree level, STEM major, city tier, months since graduation, employment status, parental tertiary education, and internship experience are controlled, these being standard determinants of early-career outcomes [15,16]. A three-item marker construct, theoretically unrelated aesthetic preference, is carried for method-bias diagnostics [65].

3.3. Analytical Strategy

The analysis proceeds in four steps. In the first step, a maximum-likelihood confirmatory factor analysis of the six substantive constructs establishes reliability, convergent and discriminant validity [66–68], measurement invariance across gender and major, and common-method diagnostics [65,69,70]. Bartlett factor scores carry the constructs into the structural step; factor-score regression of this form is consistent for the structural parameters under a correctly specified measurement model [71], and unit-weighted composites replicate every result in the robustness tests. In the second step, the following three structural equations are estimated:

$$ANX_i = a_0 + a_1 PHCD_i + a_2 SUP_i + a_3 (PHCD_i \times SUP_i) + \gamma'_1 X_i + \varepsilon_{1i} \quad (1)$$

$$ADP_i = b_0 + b_1 ANX_i + b_2 ANX_i^2 + b_3 (ANX_i \times LIT_i) + b_4 LIT_i + b_5 SUP_i + b_6 PHCD_i + \gamma'_2 X_i + \varepsilon_{2i} \quad (2)$$

$$AVD_i = c_0 + c_1 ANX_i + c_2 (ANX_i \times LIT_i) + c_3 LIT_i + c_4 SUP_i + c_5 PHCD_i + \gamma'_3 X_i + \varepsilon_{3i} \quad (3)$$

where all continuous variables are standardized so that interaction terms are interpretable in the conventional way [72], X_i denotes the control vector, and heteroskedasticity-consistent (HC3) standard errors are used throughout, with occupation-clustered inference in the robustness tests. Hypothesis H2 is not accepted from the sign of b_2 alone: following the composite testing procedure of Lind and Mehlum [32,33], a significantly positive slope at the low extreme and a significantly negative slope at the high extreme of the observed anxiety range are both required, and the turning point

$$\tau(l) = \frac{-b_1 + b_3 l}{2b_2} \quad (4)$$

is reported with a Fieller confidence interval at each level l of literacy, so that H5a is a statement about the displacement of τ . In the third step, the conditional-process analysis [30,31,73,74] computes the instantaneous indirect effect of depreciation on adaptation as the product of the conditional first-stage slope and the local second-stage derivative,

$$\omega(m, w, l) = (a_1 + a_3 w)(b_1 + 2b_2 m + b_3 l) \quad (5)$$

evaluated at chosen levels of anxiety m , support w , and literacy l , with 95 percent percentile confidence intervals from 5,000 bootstrap resamples; the indices of moderated mediation are $a_3 b_1$ for support and $a_1 b_3$ for literacy [31], and $2a_1 b_2$ indexes the curvature of the transmitted effect itself. The avoidance channel is $a_1(c_1 + c_2 l)$. In the fourth step, the fuzzy-set qualitative comparative analysis [34,35,62] calibrates the five conditions (PHCD, ANX, LIT, SUP, EXPO) and the outcome (ADP) by the direct method, with full membership, crossover, and full non-membership anchored at the 95th, 50th, and 5th sample percentiles [36]:

$$\mu(x) = \frac{1}{1 + e^{-z(x)}} \quad (6)$$

where the log-odds score is $z(x) = 3(x - c)/(f - c)$ at or above the crossover and $z(x) = 3(x - c)/(c - o)$ below it, with f , c , and o the three anchors. Necessity and sufficiency are judged by consistency and coverage:

$$Cons(X \leq Y) = \frac{\sum_i \min(x_i, y_i)}{\sum_i x_i} \quad (7)$$

$$Cov(X, Y) = \frac{\sum_i \min(x_i, y_i)}{\sum_i y_i} \quad (8)$$

with a truth-table frequency threshold of ten cases, raw consistency of at least 0.80, and proportional reduction in inconsistency (PRI) of at least 0.70 [36,63]. The intermediate solution admits only remainders consistent with the directional expectations that depreciation, literacy, support, and exposure, where present, favor adaptation, with no expectation imposed on anxiety; core conditions are those that survive into the parsimonious solution [35].

4. Results

4.1. Reliability and Validity Tests

Table 2 presents the results of the confirmatory factor analysis. The six-construct measurement model fits the data well: $\chi^2(215) = 277.35$ ($\chi^2/df = 1.29$), CFI = 0.994, TLI = 0.993, RMSEA = 0.018, SRMR = 0.022, all within the conventional cutoffs [67]. The smallest standardized loading is 0.731, the smallest Cronbach’s alpha and composite reliability are 0.79, and the smallest average variance extracted is 0.556, above the 0.50 benchmark. These results indicate satisfactory reliability and convergent validity [66].

Table 2. Constructs, items, and convergent validity.

Construct and items	Loading	α	CR	AVE
Perceived human capital depreciation (PHCD)				
phcd1	0.800	0.873	0.873	0.633
phcd2	0.780			
phcd3	0.810			
phcd4	0.791			
AI-related career anxiety (ANX)				
anx1	0.822	0.889	0.889	0.667
anx2	0.823			
anx3	0.806			
anx4	0.817			
Generative-AI literacy (LIT)				
lit1	0.795	0.865	0.865	0.616
lit2	0.790			
lit3	0.771			
lit4	0.784			
Perceived employability support (SUP)				
sup1	0.787	0.855	0.856	0.597
sup2	0.767			
sup3	0.767			
sup4	0.770			
Occupational adaptation (ADP)				
adp1	0.768	0.845	0.845	0.577
adp2	0.762			
adp3	0.757			
adp4	0.750			
Career avoidance (AVD)				
avd1	0.758	0.790	0.790	0.556
avd2	0.749			
avd3	0.731			

Note: Loadings are standardized; α is Cronbach’s alpha. Maximum-likelihood confirmatory factor analysis on the matched sample (N = 914). Model fit: $\chi^2(215) = 277.35$, CFI = 0.994, TLI = 0.993, RMSEA = 0.018, SRMR = 0.022. CR is composite reliability; AVE is average variance extracted. All loadings are significant at the 1% level. Source: authors’ own calculations.

Table 3 assesses discriminant validity from both directions. The square root of each construct’s average variance extracted (the smallest is 0.746) exceeds its largest latent correlation (the largest anywhere is 0.565), satisfying the Fornell-Larcker criterion, and the largest heterotrait-monotrait

ratio is 0.566, well below the conservative 0.85 threshold [68]. Discriminant validity is therefore established.

Table 3. Discriminant validity.

Construct	PHCD	ANX	LIT	SUP	ADP	AVD
PHCD	0.796	0.545	0.065	0.026	0.078	0.344
ANX	0.545	0.817	0.002	0.189	0.166	0.566
LIT	−0.064	−0.002	0.785	0.005	0.327	0.168
SUP	0.026	−0.190	0.004	0.773	0.211	0.219
ADP	0.079	0.166	0.327	0.212	0.760	0.022
AVD	0.343	0.565	−0.167	−0.219	−0.023	0.746

Note: Diagonal: square root of the average variance extracted. Below the diagonal: latent correlations from the confirmatory model. Above the diagonal: heterotrait-monotrait ratios. The Fornell–Larcker criterion and the HTMT < 0.85 criterion are both satisfied for every pair. Source: authors’ own calculations.

4.2. Measurement Invariance and Common Method Variance Tests

Table 4 reports the invariance and method-bias diagnostics. Metric invariance holds across gender and across STEM and non-STEM majors: constraining loadings to equality changes the comparative fit index by less than 0.001 in both cases (chi-square difference $p = 0.543$ and 0.545), so the group comparisons of Section 4.7 rest on equivalent measurement. Common method variance is bounded from three directions. By design, the outcomes were measured six months after their antecedents [69]. Statistically, the first factor of the item pool carries 22.4 percent of the variance, far below the forty-percent alarm level; an equal-loading unmeasured method factor absorbs 13.1 percent of item variance, within the range regarded as unproblematic [70]; and the largest correlation between the marker construct and any substantive score is 0.039 [65]. Method variance exists, as it always does in self-report data, but it is small, and Section 4.8 shows that partialling it out changes nothing.

Table 4. Measurement invariance and common-method diagnostics.

Model	χ^2	df	CFI	$\Delta\chi^2$ (Δdf)	p
Gender: configural	519.8	430	0.991		
Gender: metric	541.4	453	0.991	21.6 (23)	0.543
Major: configural	508.4	430	0.992		
Major: metric	530.0	453	0.992	21.6 (23)	0.545
Harman single factor: first-factor share (%)	22.4				
Method factor: variance share (%)	13.1				

Note: Configural and metric two-group models estimated by maximum likelihood; the metric model constrains loadings to equality across groups. The Harman row reports the share of item variance carried by the first principal component; the method-factor row reports the variance share of an equal-loading unmeasured method factor added to the measurement model. Source: authors’ own calculations.

4.3. Descriptive Statistics and Correlation Analysis

Table 5 reports the means, standard deviations, and correlations of the factor scores and the exposure measure. The raw correlations already outline the argument: perceived depreciation correlates with anxiety at 0.480, occupational exposure with depreciation at 0.492, and anxiety with adaptation at only 0.144 — the small linear correlation that an inverted U leaves behind

when its two arms cancel. No pairwise correlation approaches the level at which collinearity would threaten the structural estimates.

Table 5. Descriptive statistics and correlations of the factor scores.

Variable	Mean	SD	PHCD	ANX	LIT	SUP	ADP	AVD	EXPO
PHCD	0.000	1.000	1.000						
ANX	0.000	1.000	0.480	1.000					
LIT	0.000	1.000	−0.056	−0.002	1.000				
SUP	0.000	1.000	0.022	−0.166	0.003	1.000			
ADP	0.000	1.000	0.068	0.144	0.280	0.180	1.000		
AVD	0.000	1.000	0.285	0.474	−0.139	−0.180	−0.019	1.000	
EXPO	0.000	1.000	0.492	0.263	0.029	0.012	0.034	0.153	1.000

Note: Bartlett factor scores, standardized; EXPO is the standardized occupational exposure score. Lower triangle: Pearson correlations (N = 914). Correlations of magnitude 0.07 or larger are significant at the 5% level. Source: authors’ own calculations.

4.4. Hypothesis Testing: The First Stage

The first stage was investigated using the hierarchical regression method, and the results are shown in Table 6. Model (1) includes only perceived depreciation; Model (2) adds the control variables; Model (3) introduces perceived employability support; and Model (4) adds the interaction term. The results show that the regression coefficient of perceived depreciation on anxiety is 0.487 (standard error 0.027, $p < 0.01$) in the full specification, and Hypothesis H1 is verified. Support enters negatively (−0.171, $p < 0.01$), and the interaction coefficient is −0.154 ($p < 0.001$), so the conversion of depreciation into anxiety weakens as support rises; Hypothesis H6 is verified. Model (5) repeats the full specification on the complete first-wave sample (N = 1,247) with a practically identical pass-through (0.455), confirming that panel attrition does not shape the first stage.

Table 6. First stage: perceived depreciation into career anxiety (H1, H6).

	(1)	(2)	(3)	(4)	(5)
PHCD	0.480*** (0.028)	0.486*** (0.029)	0.490*** (0.028)	0.487*** (0.027)	0.455*** (0.024)
SUP			−0.170*** (0.030)	−0.171*** (0.030)	−0.171*** (0.026)
PHCD × SUP				−0.154*** (0.028)	−0.153*** (0.025)
Controls	No	Yes	Yes	Yes	Yes
Observations	914	914	914	914	1,247
R ²	0.231	0.245	0.273	0.296	0.271

Note: The dependent variable is AI-related career anxiety, standardized. Column (1) omits controls; columns (2)–(4) add controls, support, and the interaction on the matched sample; column (5) repeats column (4) on the full first-wave sample. Values in parentheses are heteroskedasticity-consistent (HC3) standard errors. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors’ own calculations.

Figure 2 visualizes the moderating effect. Evaluated one standard deviation below the mean of support, a one-standard-deviation increase in perceived depreciation raises anxiety by 0.64 standard deviations; one standard deviation above the mean, the same increase raises anxiety by only 0.33. The shaded area denotes the 95 percent confidence interval and the dashed line marks a zero marginal effect; the effect remains positive and

significant across the entire observed range of support, so support damps but does not eliminate the first stage.

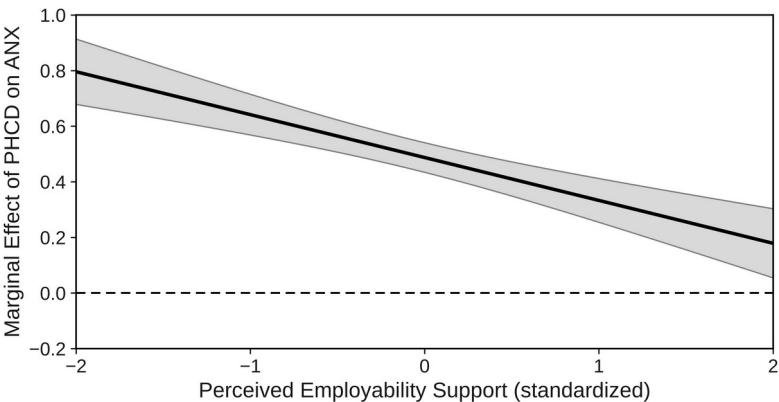


Figure 2. The moderating effect of perceived employability support on the relationship between perceived human capital depreciation and AI-related career anxiety.

4.5. Hypothesis Testing: The Curvilinear Second Stage

Table 7 estimates Equation (2) hierarchically. Model (1), which includes only the linear anxiety term, is nearly silent: anxiety appears to matter little for adaptation. Model (2) adds the squared term and the picture inverts, with the explanatory power of the model improving by 0.099 in R². Models (3) and (4) add the literacy interaction and the full antecedent set; in the full specification of Model (4) the linear coefficient is 0.174 and the quadratic coefficient is −0.237 (standard error 0.025, p < 0.01). Model (5) re-estimates Model (4) with standard errors clustered on the twenty occupation groups and the conclusion stands.

Table 7. Second stage: the curvilinear conversion of anxiety into adaptation (H2, H5a).

	(1)	(2)	(3)	(4)	(5)
ANX	0.149*** (0.037)	0.152*** (0.031)	0.153*** (0.030)	0.174*** (0.034)	0.174*** (0.030)
ANX ²		−0.266*** (0.027)	−0.256*** (0.026)	−0.237*** (0.025)	−0.237*** (0.020)
LIT			0.278*** (0.030)	0.279*** (0.029)	0.279*** (0.028)
ANX × LIT			0.138*** (0.029)	0.139*** (0.029)	0.139*** (0.028)
SUP				0.165*** (0.031)	0.165*** (0.037)
PHCD				0.009 (0.034)	0.009 (0.037)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	914	914	914	914	914
R ²	0.031	0.130	0.223	0.248	0.248

Note: The dependent variable is occupational adaptation at the second wave, standardized. Columns (1)–(4) use HC3 standard errors; column (5) repeats column (4) with standard errors

clustered on the 20 occupation groups. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

Because a negative quadratic coefficient does not by itself establish an interior optimum [32,33], Table 8 reports the composite validation. The slope at the first percentile of anxiety is 1.162 ($t = 10.31$); the slope at the 99th percentile is -0.846 ($t = -7.46$); and the Sasabuchi p -value for the composite null of monotonicity is below 0.0001. The turning point lies 0.37 standard deviations above mean anxiety, with a 95 percent Fieller interval of [0.22, 0.55] that excludes both extremes of the observed range, and 38.5 percent of respondents fall to its right. Hypothesis H2 is verified. The literacy-conditional turning points in the lower panel of Table 8 test Hypothesis H5a: the turning point moves from 0.07 standard deviations at low literacy to 0.66 at high literacy, a displacement of 0.29 per standard deviation of literacy (standard error 0.07, $p < 0.001$); a quadratic-by-literacy term is insignificant ($p = 0.434$), so literacy moves the peak without changing the curvature. Hypothesis H5a is verified.

Table 8. Formal validation of the inverted U.

Quantity	Estimate
Slope at the 1st percentile of ANX (−2.09)	1.162 ($t = 10.31$)
Slope at the 99th percentile of ANX (2.15)	−0.846 ($t = -7.46$)
Sasabuchi test of the composite null (p)	< 0.0001
Turning point (SD of ANX)	0.368 (SE 0.081)
95% Fieller interval	[0.221, 0.546]
Respondents beyond the turning point (%)	38.5
Turning point at LIT = −1 SD	0.075 (SE 0.095)
Turning point at LIT = +1 SD	0.661 (SE 0.119)
Displacement of the turning point per SD of literacy	0.293 (SE 0.067)

Note: Slopes and tests follow Lind and Mehlum's composite procedure on the full specification of Table 7, column (4); the turning point interval is a Fieller construction, and the literacy-conditional turning points and their displacement are Equation (4) evaluated on the same specification, with delta-method standard errors. Source: authors' own calculations.

Figure 3 draws the two second-stage responses on one vertical scale. Panel (a) shows the estimated adaptation response with its 95 percent confidence band; the dotted vertical line marks the turning point, and the dashed horizontal line marks the sample mean of the outcome. Panel (b) shows the avoidance response, which is monotonic throughout. The contrast between the two panels is the central descriptive fact of this study: the same anxiety that purchases correction only within a bounded band purchases escape without bound.

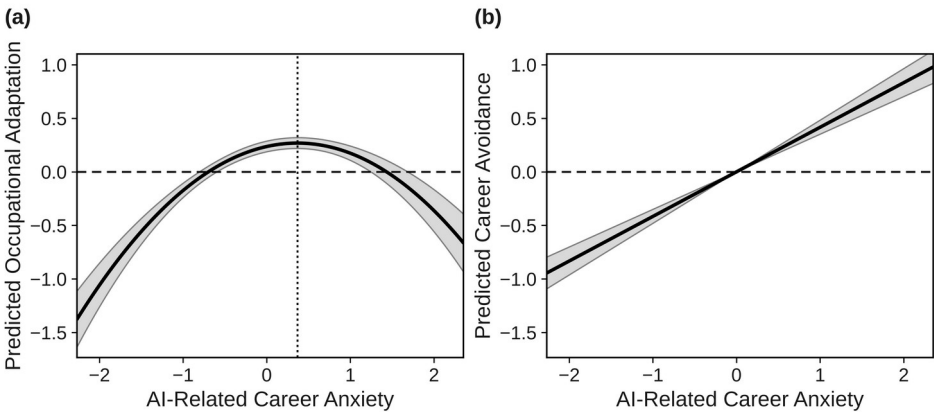


Figure 3. Predicted occupational adaptation (a) and career avoidance (b) across AI-related career anxiety.

Table 9 estimates Equation (3). The coefficient of anxiety on avoidance is 0.417 ($p < 0.01$) in the full specification, and a quadratic term added in Model (2) is small (0.043) and does not reach the five percent level ($p = 0.054$), confirming monotonicity; Hypothesis H3 is verified. The literacy interaction is negative (-0.146 , $p < 0.001$), so command of the technology drains the avoidance channel; Hypothesis H5b is verified.

Table 9. The avoidance channel (H3, H5b).

	(1)	(2)	(3)	(4)	(5)
ANX	0.472*** (0.029)	0.471*** (0.029)	0.469*** (0.028)	0.417*** (0.033)	0.417*** (0.030)
ANX ²		0.043* (0.023)			
LIT			−0.136*** (0.028)	−0.132*** (0.028)	−0.132*** (0.023)
ANX × LIT			−0.148*** (0.029)	−0.146*** (0.029)	−0.146*** (0.029)
SUP				−0.116*** (0.028)	−0.116*** (0.023)
PHCD				0.071** (0.036)	0.071* (0.036)
Controls	Yes	Yes	Yes	Yes	Yes
Observations	914	914	914	914	914
R ²	0.230	0.233	0.269	0.284	0.284

Note: The dependent variable is career avoidance at the second wave, standardized. Column (2) adds the quadratic term, which is insignificant; columns (3)–(4) add literacy and its interaction. Values in parentheses are HC3 standard errors in columns (1)–(4) and occupation-clustered standard errors in column (5). ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors’ own calculations.

4.6. Moderated Mediation Analysis

Table 10 presents the results of the bootstrapped conditional-process analysis of Equation (5). When anxiety is low and both boundary conditions are favorable, the indirect effect of perceived depreciation on adaptation through anxiety is significantly positive (0.262, with a confidence interval of [0.188, 0.340], excluding 0); when anxiety is one standard deviation above the mean

and both conditions are unfavorable, the same indirect effect is significantly negative (−0.281, [−0.371, −0.193]). The identical percept, carried by the identical channel, mobilizes the calm graduate and demobilizes the anxious one; Hypothesis H4a is verified. The index of curvilinear mediation, $2a_1b_2$, is −0.231 ([−0.286, −0.174], excluding 0), confirming that the sign reversal is a property of the transmitted effect itself. The indices of moderated mediation are −0.027 ([−0.042, −0.014]) for support in the first stage and 0.068 ([0.040, 0.096]) for literacy in the second stage [31]. The avoidance channel is uniformly positive (0.274 at low literacy and 0.132 at high literacy, both confidence intervals excluding 0), and its literacy index is −0.071 ([−0.099, −0.043]); Hypothesis H4b is verified.

Table 10. Conditional indirect effects of perceived depreciation, through anxiety (H4a, H4b).

Path and conditioning values	Estimate	95% bootstrap CI
SUP = −1, ANX = −1, LIT = −1	0.326	[0.231, 0.422]
SUP = −1, ANX = −1, LIT = +1	0.504	[0.404, 0.609]
SUP = −1, ANX = +0, LIT = −1	0.023	[−0.034, 0.078]
SUP = −1, ANX = +0, LIT = +1	0.201	[0.143, 0.260]
SUP = −1, ANX = +1, LIT = −1	−0.281	[−0.371, −0.193]
SUP = −1, ANX = +1, LIT = +1	−0.103	[−0.189, −0.015]
SUP = +1, ANX = −1, LIT = −1	0.169	[0.110, 0.235]
SUP = +1, ANX = −1, LIT = +1	0.262	[0.188, 0.340]
SUP = +1, ANX = +0, LIT = −1	0.012	[−0.018, 0.041]
SUP = +1, ANX = +0, LIT = +1	0.104	[0.069, 0.143]
SUP = +1, ANX = +1, LIT = −1	−0.146	[−0.205, −0.094]
SUP = +1, ANX = +1, LIT = +1	−0.053	[−0.103, −0.008]
Avoidance path, LIT = −1	0.274	[0.226, 0.325]
Avoidance path, LIT = +1	0.132	[0.090, 0.176]
IMM of SUP (first stage, adaptation)	−0.027	[−0.042, −0.014]
IMM of LIT (second stage, adaptation)	0.068	[0.040, 0.096]
Index of curvilinear mediation (2·a1·b2)	−0.231	[−0.286, −0.174]
IMM of LIT (second stage, avoidance)	−0.071	[−0.099, −0.043]

Note: Instantaneous indirect effects $\omega(m, w, l)$ of Equation (5), with SUP, ANX and LIT set to the stated values in standard deviations. Confidence intervals are percentile intervals from 5,000 bootstrap resamples of individuals; the seed is fixed in the released code. IMM denotes an index of moderated mediation. Source: authors’ own calculations.

4.7. Heterogeneity Analysis

Table 11 re-estimates the full adaptation model within ten subsamples defined by gender, major, city tier, occupational exposure, and employment status. The quadratic coefficient is negative and significant in every subsample, and none of the five between-group differences in the marginal effect of anxiety approaches statistical significance (the smallest p-value is 0.492). The inverted U is therefore not carried by any single stratum of the cohort; it is a general property of the population studied, which is consistent with the interpretation of saturation as a structural feature of the control loop rather than an artifact of pooling heterogeneous linear responses.

Table 11. Subsample estimates: the mechanism across the cohort.

Subsample	N	dADP/dANX at mean	(SE)	ANX ²
Female	483	0.187	(0.048)	−0.260***
Male	431	0.166	(0.052)	−0.205***
difference (z, p)		0.29	0.774	
STEM	368	0.200	(0.055)	−0.228***
Non-STEM	546	0.161	(0.046)	−0.244***
difference (z, p)		0.55	0.586	
Tier-1 city	298	0.209	(0.062)	−0.240***
Other cities	616	0.158	(0.042)	−0.231***
difference (z, p)		0.69	0.492	
High exposure	477	0.180	(0.046)	−0.213***
Low exposure	437	0.164	(0.056)	−0.255***
difference (z, p)		0.21	0.831	
Employed	676	0.181	(0.038)	−0.238***
Not employed	238	0.171	(0.083)	−0.224***
difference (z, p)		0.11	0.914	

Note: Each pair of rows re-estimates the full adaptation model within the stated subsample; the third column is the marginal effect of anxiety at the sample mean, the final column the subsample quadratic coefficient. Difference rows report a two-sample z-test of the marginal effects. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.8. Robustness and Attrition Tests

Table 12 subjects the curvilinear second stage to six replications. (1) Replacing the adaptation index with the log of reported reskilling hours, a behavioral count rather than a Likert judgment, returns a quadratic of −0.140 ($p < 0.01$). (2) Weighting the retained sample by inverse retention probabilities estimated from all first-wave observables leaves the quadratic at −0.237. (3) Excluding the software and information-technology occupations gives −0.235. (4) Unit-weighted composites in place of factor scores give −0.237 [71]. (5) Partialling the marker score out of every construct gives −0.240 [65]. (6) The wild-cluster bootstrap over the twenty occupation groups, the appropriate small-sample inference [75], puts the p-value of the quadratic term at 0.001. The inverted U survives every change of measure, sample, weighting, and inference.

Table 12. Robustness of the curvilinear second stage.

Specification	ANX	(SE)	ANX ²	(SE)	N
(1) log reskilling hours	0.090***	(0.029)	−0.140***	(0.022)	914
(2) inverse-probability weighted	0.174***	(0.029)	−0.237***	(0.021)	914
(3) excluding software and IT	0.170***	(0.035)	−0.235***	(0.027)	838
(4) unit-weighted composites	0.174***	(0.034)	−0.237***	(0.025)	914
(5) marker-partialled scores	0.178***	(0.034)	−0.240***	(0.025)	914
(6) wild cluster bootstrap p for ANX ² (20 occupation clusters)			0.001		914

Note: Each row replicates the full adaptation specification under the stated change. Row (2) weights by inverse retention probabilities from a first-wave linear probability model; row (6) reports the wild-cluster bootstrap p-value for the quadratic term with Rademacher weights and

the null imposed, 999 replications. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.9. Configurational Analysis

Table 13 reports the calibration anchors and the necessity analysis. No condition, present or absent, reaches the 0.90 consistency conventionally required of a necessary condition; the maximum anywhere is 0.708. Hypothesis H7a is verified: there is no single gate through which every adapting graduate passes.

Table 13. Fuzzy-set calibration and necessity analysis.

Condition	Full (0.95)	Crossover (0.50)	Out (0.05)
Panel A: calibration anchors (raw score at membership 0.95 / 0.50 / 0.05)			
PHCD	1.70	0.03	−1.75
ANX	1.58	0.03	−1.68
LIT	1.63	0.03	−1.76
SUP	1.64	0.01	−1.61
EXPO	1.56	0.09	−1.89
Panel B: necessity analysis for high adaptation			
Condition	Consistency	Coverage	
PHCD	0.665	0.667	
~PHCD	0.639	0.647	
ANX	0.674	0.677	
~ANX	0.614	0.621	
LIT	0.708	0.718	
~LIT	0.592	0.593	
SUP	0.682	0.681	
~SUP	0.605	0.615	
EXPO	0.645	0.659	
~EXPO	0.647	0.643	

Note: Direct-method calibration with anchors at the 95th, 50th and 5th sample percentiles of each standardized score. ~ denotes negation. No consistency reaches the 0.90 threshold conventionally required of a necessary condition. Source: authors' own calculations.

Sufficiency is a different matter. Of the 31 truth-table rows that clear the frequency threshold, only 4 pass both consistency screens, and they minimize to a single intermediate solution, identical to the parsimonious solution, so every condition in it is core [35]: the conjunction of anxiety, literacy, and support (Table 14), with consistency 0.892 and coverage 0.406. The near-misses are informative: configurations combining anxiety and literacy without support reach raw consistencies near 0.91 yet fail the proportional-reduction screen, and the highest-consistency row for the negation of adaptation (raw consistency 0.893) fails the same screen ($PRI\ 0.602 < 0.70$), so no configuration is reliably sufficient for failing to adapt. Hypothesis H7b is therefore not supported, and its failure is itself a finding: the system offers one road to high adaptation, not many. Anxiety appears in the recipe — the signal is an ingredient of correction — but it never suffices without the actuator and the environment, which is precisely the conjunction the control-loop framing predicted [27,29].

Table 14. Sufficient configuration for high occupational adaptation.

	PHCD	ANX	LIT	SUP	EXPO	Consistency	Raw cov.	Unique cov.
C1		●	●	●		0.892	0.406	0.406
Solution						0.892	0.406	

Note: ● indicates the presence of a condition and ⊗ its absence; a blank cell means the condition is irrelevant to the configuration. All marked conditions are core: the parsimonious and intermediate solutions coincide. Truth-table thresholds: frequency ≥ 10 cases, raw consistency ≥ 0.80, PRI ≥ 0.70. No configuration passes the same thresholds for the negation of adaptation. Source: authors’ own calculations.

5. Conclusions

5.1. Research Conclusions

Based on socio-technical systems theory and a cybernetic control perspective, this study analyzes how perceived human capital depreciation shapes the early-career choices of fresh graduates in the era of generative artificial intelligence, using two-wave survey data from Chinese graduates within three years of graduation. The conclusions are as follows: (1) perceived human capital depreciation significantly raises AI-related career anxiety, and perceived employability support buffers this conversion, cutting the pass-through from 0.64 to 0.33; (2) career anxiety converts into occupational adaptation along an inverted U-shaped path with a turning point 0.37 standard deviations above mean anxiety — 38.5 percent of the cohort already stands beyond it — while career avoidance rises monotonically; (3) generative-AI literacy displaces the turning point outward from 0.07 to 0.66 standard deviations and weakens the avoidance channel, and the indirect effect of depreciation through anxiety reverses sign across the anxiety distribution; and (4) exactly one configuration — anxiety combined with literacy and support — is sufficient for high adaptation, no single condition is necessary, and no configuration is sufficient for the negation.

5.2. Theoretical Contributions

The main theoretical contributions of this paper are as follows: First, this paper extends the study of generative artificial intelligence and work from the demand side to the supply side of the entry labor market. The existing literature has measured how adoption reshapes tasks, hiring, and productivity [1,4,6,76], while the perception-to-behavior chain inside the displaced margin has remained unexamined. By showing that the behavioral response of the entry cohort is nonmonotonic — mobilization up to a turning point that two-fifths of the cohort has already passed, demobilization beyond it — this study supplies a psychological microfoundation for early-career scarring [15,18] and cautions against adjustment models that assume displacement pressure translates smoothly into retraining uptake [13,14].

Second, this paper contributes a systems treatment to the artificial-intelligence anxiety literature. Prior studies have treated anxiety mainly as an outcome to be predicted or a hazard to be reduced [19,20,22]. Modeling anxiety instead as the error signal of a cybernetic feedback loop [27–29] yields three predictions that a valence-only reading does not — an interior optimum, actuator-dependent displacement of that optimum, and conjunctural sufficiency — and all three survived their tests. The finding connects the psychology of technological threat to the organizational learning literature on

absorption capacity [77–79]: signals without the capacity to act on them produce noise, not learning.

Third, this paper demonstrates the value of combining conditional-process and configurational methods within one design. The regression apparatus locates the turning point and its displacement; the fuzzy-set analysis reveals that the conditions operate as a conjunction rather than as additive levers, and that equifinality — the default expectation of configurational research [35,36] — fails here in a theoretically informative way. Reporting that failure as a finding, rather than forcing a multi-recipe reading, is the faithful use of the method [62,63] and shows how a tight systems theory can predict, and data can deliver, a single path.

5.3. Management Implications

The empirical results generate actionable implications for the three actors who govern the graduate transition. For universities, the binding constraints are the actuator and the environment, not the volume of the alarm. Career services that concentrate on reassurance address the sensor; embedding supervised, verifiable generative-AI practice into every curriculum [7,23] buys a double dividend, raising adaptation directly and extending the anxiety range over which graduates keep adapting rather than freezing, while visible, credible transition infrastructure — placement pathways, employer-linked reskilling, alumni networks that answer — lowers the anxiety produced by any given level of objective threat.

For employers currently thinning their junior pipelines [4,8], the configurational result supplies a one-line design rule: anxiety converts into adaptation only in the presence of literacy and support, so any communication of threat should arrive bundled with tool access, training time, and a visible internal pathway [80,81]. A firm that hires fewer entrants and tells the remainder to upskill or leave is raising the error signal while shrinking the actuator set — the exact combination the estimates identify as producing avoidance and exit. For policymakers confronting record graduate cohorts [10,82–84], the estimates argue for shifting marginal spending from awareness campaigns, which raise the signal, toward subsidized generative-AI credentialing and employer-linked placement, which complete the loop — and for targeting that spending on the high-anxiety segment, where the same expenditure buys the largest swing from demobilization back to adaptation.

5.4. Limitations and Future Directions

This study has several limitations that point to future research. First, although the outcomes are separated from their antecedents by six months and the attrition diagnostics are clean, the design remains observational; unmeasured traits such as dispositional neuroticism could raise both anxiety and behavioral reports, and only a design that manipulates the actuators — for instance, randomized literacy training layered on measured anxiety — can fully close that door. Second, all psychological constructs are self-reported; the behavioral hours measure and the method-bias diagnostics bound the resulting bias but do not eliminate it [70]. Third, the sample consists of Chinese graduates observed at one moment of one technological transition; the location of the turning point is surely context-dependent, and comparative replication across institutional regimes with different transition infrastructures [82,85] is a natural next step. Fourth, the occupational exposure score is an occupation-level construct assigned to individuals; task-

level exposure measured within jobs would sharpen both the regression and the calibration. Fifth, the two-wave design identifies one traversal of the loop, not its dynamics; whether adaptation feeds back to lower depreciation perceptions, closing a balancing loop, or whether avoidance compounds into the reinforcing spiral that conservation of resources theory predicts [48,86–90], requires a multi-wave panel analyzed with the estimation machinery the system-dynamics tradition has built for such loops [91]. Finally, richer systems lenses — soft systems inquiry into how graduates collectively make sense of the transition [92,93], and comparisons with adjacent evidence on technology and employment in China [3,5,57,94–104] — would further test the generality of the loop identified here.

Author Contributions: Conceptualization, X.C. and T.M.; methodology, X.C.; software, X.C.; validation, T.M.; formal analysis, X.C.; investigation, T.M.; data curation, T.M.; writing—original draft preparation, X.C.; writing—review and editing, T.M.; visualization, X.C.; supervision, T.M.; project administration, T.M. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: The study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board of the College of Business, Jiaying University (protocol code JXU-BUS-2025-014, approved 22 August 2025).

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The data presented in this study are available on reasonable request from the corresponding author. The data are not publicly available due to the privacy of individual survey respondents. The complete estimation code that produces every table and figure is available from the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Eloundou, T.; Manning, S.; Mishkin, P.; Rock, D. GPTs Are GPTs: Labor Market Impact Potential of LLMs. *Science* **2024**, *384*, 1306–1308. [CrossRef]
2. Felten, E.; Raj, M.; Seamans, R. Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses. *Strateg. Manag. J.* **2021**, *42*, 2195–2217. [CrossRef]
3. Noy, S.; Zhang, W. Experimental Evidence on the Productivity Effects of Generative Artificial Intelligence. *Science* **2023**, *381*, 187–192. [CrossRef]
4. Brynjolfsson, E.; Chandar, B.; Chen, R. *Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence*; Working Paper; Stanford Digital Economy Lab, Stanford University: Stanford, CA, USA, 2025.
5. Hui, X.; Reshef, O.; Zhou, L. The Short-Term Effects of Generative Artificial Intelligence on Employment: Evidence from an Online Labor Market. *Organ. Sci.* **2024**, *35*, 1977–1989. [CrossRef]
6. Brynjolfsson, E.; Li, D.; Raymond, L.R. Generative AI at Work. *Q. J. Econ.* **2025**, *140*, 889–942. [CrossRef]
7. Dell’Acqua, F.; McFowland, E., III; Mollick, E.R.; Lifshitz-Assaf, H.; Kellogg, K.C.; Rajendran, S.; Kraye, L.; Candelon, F.; Lakhani, K.R. Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of Artificial Intelligence on Knowledge Worker Productivity and Quality. *Organ. Sci.* **2026**, in press. [CrossRef]
8. Autor, D. *Applying AI to Rebuild Middle Class Jobs*; NBER Working Paper 32140; National Bureau of Economic Research: Cambridge, MA, USA, 2024.
9. Autor, D.; Chin, C.; Salomons, A.; Seegmiller, B. New Frontiers: The Origins and Content of New Work, 1940–2018. *Q. J. Econ.* **2024**, *139*, 1399–1465. [CrossRef]

10. Ministry of Education of the People's Republic of China. The Number of College Graduates Nationwide Is Expected to Reach 12.22 Million in 2025. Available online: <http://www.moe.gov.cn/> (accessed on 20 March 2026).
11. Li, S.; Whalley, J.; Xing, C. China's Higher Education Expansion and Unemployment of College Graduates. *China Econ. Rev.* **2014**, *30*, 567–582. [CrossRef]
12. Acemoglu, D.; Restrepo, P. Tasks, Automation, and the Rise in U.S. Wage Inequality. *Econometrica* **2022**, *90*, 1973–2016. [CrossRef]
13. Autor, D.H. Why Are There Still So Many Jobs? The History and Future of Workplace Automation. *J. Econ. Perspect.* **2015**, *29*, 3–30. [CrossRef]
14. Acemoglu, D.; Restrepo, P. The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment. *Am. Econ. Rev.* **2018**, *108*, 1488–1542. [CrossRef]
15. Kahn, L.B. The Long-Term Labor Market Consequences of Graduating from College in a Bad Economy. *Labour Econ.* **2010**, *17*, 303–316. [CrossRef]
16. Oreopoulos, P.; von Wachter, T.; Heisz, A. The Short- and Long-Term Career Effects of Graduating in a Recession. *Am. Econ. J. Appl. Econ.* **2012**, *4*, 1–29. [CrossRef]
17. Schwandt, H.; von Wachter, T. Unlucky Cohorts: Estimating the Long-Term Effects of Entering the Labor Market in a Recession in Large Cross-Sectional Data Sets. *J. Labor Econ.* **2019**, *37*, S161–S198. [CrossRef]
18. von Wachter, T. The Persistent Effects of Initial Labor Market Conditions for Young Adults and Their Sources. *J. Econ. Perspect.* **2020**, *34*, 168–194. [CrossRef]
19. Wang, Y.-Y.; Wang, Y.-S. Development and Validation of an Artificial Intelligence Anxiety Scale: An Initial Application in Predicting Motivated Learning Behavior. *Interact. Learn. Environ.* **2022**, *30*, 619–634. [CrossRef]
20. Kong, H.; Yuan, Y.; Baruch, Y.; Bu, N.; Jiang, X.; Wang, K. Influences of Artificial Intelligence (AI) Awareness on Career Competency and Job Burnout. *Int. J. Contemp. Hosp. Manag.* **2021**, *33*, 717–734. [CrossRef]
21. Brougham, D.; Haar, J. Smart Technology, Artificial Intelligence, Robotics, and Algorithms (STARA): Employees' Perceptions of Our Future Workplace. *J. Manag. Organ.* **2018**, *24*, 239–257. [CrossRef]
22. Tarafdar, M.; Tu, Q.; Ragu-Nathan, B.S.; Ragu-Nathan, T.S. The Impact of Technostress on Role Stress and Productivity. *J. Manag. Inf. Syst.* **2007**, *24*, 301–328. [CrossRef]
23. Chan, C.K.Y.; Hu, W. Students' Voices on Generative AI: Perceptions, Benefits, and Challenges in Higher Education. *Int. J. Educ. Technol. High. Educ.* **2023**, *20*, 43. [CrossRef]
24. Trist, E.L.; Bamforth, K.W. Some Social and Psychological Consequences of the Longwall Method of Coal-Getting. *Hum. Relat.* **1951**, *4*, 3–38. [CrossRef]
25. Emery, F.E.; Trist, E.L. The Causal Texture of Organizational Environments. *Hum. Relat.* **1965**, *18*, 21–32. [CrossRef]
26. von Bertalanffy, L. *General System Theory: Foundations, Development, Applications*; George Braziller: New York, NY, USA, 1968.
27. Ashby, W.R. *An Introduction to Cybernetics*; Chapman and Hall: London, UK, 1956.
28. Beer, S. *Brain of the Firm*, 2nd ed.; Wiley: Chichester, UK, 1981.
29. Carver, C.S.; Scheier, M.F. Control Theory: A Useful Conceptual Framework for Personality-Social, Clinical, and Health Psychology. *Psychol. Bull.* **1982**, *92*, 111–135. [CrossRef]
30. Edwards, J.R.; Lambert, L.S. Methods for Integrating Moderation and Mediation: A General Analytical Framework Using Moderated Path Analysis. *Psychol. Methods* **2007**, *12*, 1–22. [CrossRef]
31. Hayes, A.F. An Index and Test of Linear Moderated Mediation. *Multivariate Behav. Res.* **2015**, *50*, 1–22. [CrossRef]
32. Lind, J.T.; Mehlum, H. With or Without U? The Appropriate Test for a U-Shaped Relationship. *Oxf. Bull. Econ. Stat.* **2010**, *72*, 109–118. [CrossRef]
33. Haans, R.F.J.; Pieters, C.; He, Z.-L. Thinking about U: Theorizing and Testing U- and Inverted U-Shaped Relationships in Strategy Research. *Strateg. Manag. J.* **2016**, *37*, 1177–1195. [CrossRef]
34. Ragin, C.C. *Redesigning Social Inquiry: Fuzzy Sets and Beyond*; University of Chicago Press: Chicago, IL, USA, 2008.
35. Fiss, P.C. Building Better Causal Theories: A Fuzzy Set Approach to Typologies in Organization Research. *Acad. Manag. J.* **2011**, *54*, 393–420. [CrossRef]
36. Pappas, I.O.; Woodside, A.G. Fuzzy-Set Qualitative Comparative Analysis (fsQCA): Guidelines for Research Practice in Information Systems and Marketing. *Int. J. Inf. Manag.* **2021**, *58*, 102310. [CrossRef]
37. Dwivedi, Y.K.; Kshetri, N.; Hughes, L.; Slade, E.L.; Jeyaraj, A.; Kar, A.K.; Baabdullah, A.M.; Koohang, A.; Raghavan, V.; Ahuja, M.; et al. Opinion Paper: “So What If ChatGPT Wrote It?” Multidisciplinary Perspectives on Opportunities, Challenges and Implications of Generative Conversational AI for Research, Practice and Policy. *Int. J. Inf. Manag.* **2023**, *71*, 102642. [CrossRef]
38. Janssen, M. Responsible Governance of Generative AI: Conceptualizing GenAI as Complex Adaptive Systems. *Policy Soc.* **2025**, *44*, 38–51. [CrossRef]
39. Becker, G.S. *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*; National Bureau of Economic Research: New York, NY, USA, 1964.

40. De Grip, A.; Van Loo, J. The Economics of Skills Obsolescence: A Review. In *The Economics of Skills Obsolescence (Research in Labor Economics, Vol. 21)*; De Grip, A.; Van Loo, J.; Mayhew, K.; Emerald: Bingley, UK, 2002; 1–26.
41. Deming, D.J.; Noray, K. Earnings Dynamics, Changing Job Skills, and STEM Careers. *Q. J. Econ.* **2020**, *135*, 1965–2005. [CrossRef]
42. Arrow, K.J. The Economic Implications of Learning by Doing. *Rev. Econ. Stud.* **1962**, *29*, 155–173. [CrossRef]
43. Lazear, E.P. Firm-Specific Human Capital: A Skill-Weights Approach. *J. Political Econ.* **2009**, *117*, 914–940. [CrossRef]
44. Gibbons, R.; Waldman, M. Task-Specific Human Capital. *Am. Econ. Rev.* **2004**, *94*, 203–207. [CrossRef]
45. Autor, D.H.; Levy, F.; Murnane, R.J. The Skill Content of Recent Technological Change: An Empirical Exploration. *Q. J. Econ.* **2003**, *118*, 1279–1333. [CrossRef]
46. Lazarus, R.S.; Folkman, S. *Stress, Appraisal, and Coping*; Springer: New York, NY, USA, 1984.
47. Hobfoll, S.E. Conservation of Resources: A New Attempt at Conceptualizing Stress. *Am. Psychol.* **1989**, *44*, 513–524. [CrossRef]
48. Hobfoll, S.E.; Halbesleben, J.; Neveu, J.-P.; Westman, M. Conservation of Resources in the Organizational Context: The Reality of Resources and Their Consequences. *Annu. Rev. Organ. Psychol. Organ. Behav.* **2018**, *5*, 103–128. [CrossRef]
49. Pisarik, C.T.; Rowell, P.C.; Thompson, L.K. A Phenomenological Study of Career Anxiety among College Students. *Career Dev. Q.* **2017**, *65*, 339–352. [CrossRef]
50. Savickas, M.L.; Porfeli, E.J. Career Adapt-Abilities Scale: Construction, Reliability, and Measurement Equivalence across 13 Countries. *J. Vocat. Behav.* **2012**, *80*, 661–673. [CrossRef]
51. Lent, R.W.; Brown, S.D. Social Cognitive Model of Career Self-Management: Toward a Unifying View of Adaptive Career Behavior across the Life Span. *J. Couns. Psychol.* **2013**, *60*, 557–568. [CrossRef]
52. Yerkes, R.M.; Dodson, J.D. The Relation of Strength of Stimulus to Rapidity of Habit-Formation. *J. Comp. Neurol. Psychol.* **1908**, *18*, 459–482. [CrossRef]
53. Grant, A.M.; Schwartz, B. Too Much of a Good Thing: The Challenge and Opportunity of the Inverted U. *Perspect. Psychol. Sci.* **2011**, *6*, 61–76. [CrossRef]
54. Pierce, J.R.; Aguinis, H. The Too-Much-of-a-Good-Thing Effect in Management. *J. Manag.* **2013**, *39*, 313–338. [CrossRef]
55. Bandura, A. *Self-Efficacy: The Exercise of Control*; W.H. Freeman: New York, NY, USA, 1997.
56. Raisch, S.; Krakowski, S. Artificial Intelligence and Management: The Automation-Augmentation Paradox. *Acad. Manag. Rev.* **2021**, *46*, 192–210. [CrossRef]
57. Vaccaro, M.; Almaatouq, A.; Malone, T. When Combinations of Humans and AI Are Useful: A Systematic Review and Meta-Analysis. *Nat. Hum. Behav.* **2024**, *8*, 2293–2303. [CrossRef]
58. Fugate, M.; Kinicki, A.J.; Ashforth, B.E. Employability: A Psycho-Social Construct, Its Dimensions, and Applications. *J. Vocat. Behav.* **2004**, *65*, 14–38. [CrossRef]
59. Akkermans, J.; Richardson, J.; Kraimer, M.L. The Covid-19 Crisis as a Career Shock: Implications for Careers and Vocational Behavior. *J. Vocat. Behav.* **2020**, *119*, 103434. [CrossRef]
60. Simon, H.A. The Architecture of Complexity. *Proc. Am. Philos. Soc.* **1962**, *106*, 467–482.
61. Holland, J.H. *Hidden Order: How Adaptation Builds Complexity*; Addison-Wesley: Reading, MA, USA, 1995.
62. Schneider, C.Q.; Wagemann, C. *Set-Theoretic Methods for the Social Sciences: A Guide to Qualitative Comparative Analysis*; Cambridge University Press: Cambridge, UK, 2012.
63. Greckhamer, T.; Furnari, S.; Fiss, P.C.; Aguilera, R.V. Studying Configurations with Qualitative Comparative Analysis: Best Practices in Strategy and Organization Research. *Strateg. Organ.* **2018**, *16*, 482–495. [CrossRef]
64. National Bureau of Statistics of China. Surveyed Urban Unemployment Rate of the Population Aged 16 to 24 Excluding Students. Available online: <https://www.stats.gov.cn/english/> (accessed on 20 March 2026).
65. Lindell, M.K.; Whitney, D.J. Accounting for Common Method Variance in Cross-Sectional Research Designs. *J. Appl. Psychol.* **2001**, *86*, 114–121. [CrossRef]
66. Fornell, C.; Larcker, D.F. Evaluating Structural Equation Models with Unobservable Variables and Measurement Error. *J. Mark. Res.* **1981**, *18*, 39–50. [CrossRef]
67. Hu, L.; Bentler, P.M. Cutoff Criteria for Fit Indexes in Covariance Structure Analysis: Conventional Criteria versus New Alternatives. *Struct. Equ. Model.* **1999**, *6*, 1–55. [CrossRef]
68. Henseler, J.; Ringle, C.M.; Sarstedt, M. A New Criterion for Assessing Discriminant Validity in Variance-Based Structural Equation Modeling. *J. Acad. Mark. Sci.* **2015**, *43*, 115–135. [CrossRef]
69. Podsakoff, P.M.; MacKenzie, S.B.; Lee, J.-Y.; Podsakoff, N.P. Common Method Biases in Behavioral Research: A Critical Review of the Literature and Recommended Remedies. *J. Appl. Psychol.* **2003**, *88*, 879–903. [CrossRef]
70. Podsakoff, P.M.; Podsakoff, N.P.; Williams, L.J.; Huang, C.; Yang, J. Common Method Bias: It's Bad, It's Complex, It's Widespread, and It's Not Easy to Fix. *Annu. Rev. Organ. Psychol. Organ. Behav.* **2024**, *11*, 17–61. [CrossRef]
71. Devlieger, I.; Mayer, A.; Rosseel, Y. Hypothesis Testing Using Factor Score Regression: A Comparison of Four Methods. *Educ. Psychol. Meas.* **2016**, *76*, 741–770. [CrossRef]

72. Aiken, L.S.; West, S.G. *Multiple Regression: Testing and Interpreting Interactions*; Sage: Newbury Park, CA, USA, 1991.
73. Preacher, K.J.; Hayes, A.F. Asymptotic and Resampling Strategies for Assessing and Comparing Indirect Effects in Multiple Mediator Models. *Behav. Res. Methods* **2008**, *40*, 879–891. [CrossRef]
74. Zhao, X.; Lynch, J.G., Jr.; Chen, Q. Reconsidering Baron and Kenny: Myths and Truths about Mediation Analysis. *J. Consum. Res.* **2010**, *37*, 197–206. [CrossRef]
75. Cameron, A.C.; Gelbach, J.B.; Miller, D.L. Bootstrap-Based Improvements for Inference with Clustered Errors. *Rev. Econ. Stat.* **2008**, *90*, 414–427. [CrossRef]
76. Babina, T.; Fedyk, A.; He, A.; Hodson, J. Artificial Intelligence, Firm Growth, and Product Innovation. *J. Financ. Econ.* **2024**, *151*, 103745. [CrossRef]
77. Cohen, W.M.; Levinthal, D.A. Absorptive Capacity: A New Perspective on Learning and Innovation. *Adm. Sci. Q.* **1990**, *35*, 128–152. [CrossRef]
78. March, J.G. Exploration and Exploitation in Organizational Learning. *Organ. Sci.* **1991**, *2*, 71–87. [CrossRef]
79. Argote, L.; Miron-Spektor, E. Organizational Learning: From Experience to Knowledge. *Organ. Sci.* **2011**, *22*, 1123–1137. [CrossRef]
80. Kellogg, K.C.; Valentine, M.A.; Christin, A. Algorithms at Work: The New Contested Terrain of Control. *Acad. Manag. Ann.* **2020**, *14*, 366–410. [CrossRef]
81. Beane, M. Shadow Learning: Building Robotic Surgical Skill When Approved Means Fail. *Adm. Sci. Q.* **2019**, *64*, 87–123. [CrossRef]
82. International Labour Organization. *Global Employment Trends for Youth 2024: Decent Work, Brighter Futures*; International Labour Office: Geneva, Switzerland, 2024.
83. International Labour Organization. *World Employment and Social Outlook: Trends 2025*; International Labour Office: Geneva, Switzerland, 2025.
84. State Council of the People's Republic of China. Opinions on Deepening the Implementation of the "Artificial Intelligence Plus" Initiative. Available online: <https://www.gov.cn/> (accessed on 20 March 2026).
85. World Economic Forum. *The Future of Jobs Report 2025*; World Economic Forum: Geneva, Switzerland, 2025.
86. Sterman, J.D. *Business Dynamics: Systems Thinking and Modeling for a Complex World*; Irwin/McGraw-Hill: Boston, MA, USA, 2000.
87. Forrester, J.W. *Industrial Dynamics*; MIT Press: Cambridge, MA, USA, 1961.
88. Meadows, D.H. *Thinking in Systems: A Primer*; Chelsea Green Publishing: White River Junction, VT, USA, 2008.
89. Senge, P.M. *The Fifth Discipline: The Art and Practice of the Learning Organization*; Doubleday: New York, NY, USA, 1990.
90. Repenning, N.P.; Sterman, J.D. Capability Traps and Self-Confirming Attribution Errors in the Dynamics of Process Improvement. *Adm. Sci. Q.* **2002**, *47*, 265–295. [CrossRef]
91. Rahmandad, H.; Repenning, N. Capability Erosion Dynamics. *Strateg. Manag. J.* **2016**, *37*, 649–672. [CrossRef]
92. Checkland, P. *Systems Thinking, Systems Practice*; Wiley: Chichester, UK, 1981.
93. Weick, K.E. The Collapse of Sensemaking in Organizations: The Mann Gulch Disaster. *Adm. Sci. Q.* **1993**, *38*, 628–652. [CrossRef]
94. Liang, H.; Fan, J.; Wang, Y. Artificial Intelligence, Technological Innovation, and Employment Transformation for Sustainable Development: Evidence from China. *Sustainability* **2025**, *17*, 3842. [CrossRef]
95. Xue, X.; Li, L.; Chen, J.; Luo, T. How Does Digital Technology Innovation Quality Empower Corporate ESG Performance? The Roles of Digital Transformation and Digital Technology Diffusion. *Systems* **2025**, *13*, 929. [CrossRef]
96. Xue, F.; Jin, S. Artificial Intelligence Adoption, Innovation Efficiency, and Governance Mechanisms: Evidence from China. *Systems* **2025**, *13*, 1062. [CrossRef]
97. Yang, B.; Sun, Y.; Zeng, Z.; Li, Q. Deskilling, Reskilling, or Upskilling? Unpacking the Pathways of Student Adaptation to Generative Artificial Intelligence. *Int. J. Inf. Manag.* **2026**, *87*, 103002. [CrossRef]
98. Machucho, R.; Ortiz, D. The Impacts of Artificial Intelligence on Business Innovation: A Comprehensive Review of Applications, Organizational Challenges, and Ethical Considerations. *Systems* **2025**, *13*, 264. [CrossRef]
99. Humlum, A.; Vestergaard, E. *Large Language Models, Small Labor Market Effects*; NBER Working Paper 33777; National Bureau of Economic Research: Cambridge, MA, USA, 2025.
100. Otis, N.G.; Clarke, R.; Delecourt, S.; Holtz, D.; Koning, R. The Uneven Impact of Generative Artificial Intelligence on Entrepreneurial Performance: Evidence from a Field Experiment in Kenya. *Manag. Sci.* **2026**, in press. [CrossRef]
101. Forsythe, E. Why Don't Firms Hire Young Workers during Recessions? *Econ. J.* **2022**, *132*, 1765–1789. [CrossRef]
102. Cui, Z.K.; Demirer, M.; Jaffe, S.; Musolff, L.; Peng, S.; Salz, T. The Effects of Generative AI on High-Skilled Work: Evidence from Three Field Experiments with Software Developers. *Manag. Sci.* **2026**, in press. [CrossRef]
103. Peng, S.; Kalliamvakou, E.; Cihon, P.; Demirer, M. The Impact of AI on Developer Productivity: Evidence from GitHub Copilot. *arXiv* **2023**, arXiv:2302.06590.

936 104. Bick, A.; Blandin, A.; Deming, D.J. *The Rapid Adoption of Generative AI*; NBER Working Paper 32966; National
937 Bureau of Economic Research: Cambridge, MA, USA, 2024.